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STA 321 — Past Questions & Materials

by Renance

Course: STA 321

Institution: Delta State Polytechnic, Otefe-Oghara

Papers in this pack: 2

Levels: HND1

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Paper 1 of 2 — STATISTICAL THEORY IV STA 321 HND1 (20232024) SECOND SEMESTER

Level HND1 • First Semester • Handout

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DELTA STATE POLYTECHNIC, OTEFE-OGHARA
SCHOOL OF APPLIED SCIENCES AND TECHNOLOGY
DEPARTMENT OF STATISTICS

SECOND SEMESTER EXAMINATION 2023/2024 CADEMIC SESSION

COURSE TITLE: STATISTICAL THEORY IV
COURSE CODE: STA 321
PROGRAMME OF STUDY: HNDI COMPUTER SCIENCE & STATISTICS
TIME: 2½ HOURS
INSTRUCTION: ATTEMPT ANY FIVE (5) QUESTIONS

Question One:

- 1a. Given the Chi-square distribution with degrees of freedom as the mean of 20, 30, 40, 60, find:
(i) The Mean (ii) Median (iii) Variance (iv) Skewness (v) Mode
- b. Let X be a random variable, define the characteristic function of;
(i) the discrete case (ii) The continuous case
- c. Outline the basic steps in hypothesis testing

Question Two:

- 2a. 200 is the mean of a population of 100 items which variance is 49. Miss Gift the Course Rep. was asked to select 26 items randomly from the population and she must not drop it back after each selection. The mean of this experiment is 196. At 1% level of significance, has the population mean of the items decreased?
- b. (i) State any two uses of F-test
(ii) If m and n is the degree of freedom of a F-distribution, find the;
(a) Mean and (b) Variance of the distribution, given m = 10, and n = 8

Question Three:

- 3a. If X has a gamma distribution with parameters, r and λ ,
(i) Show that the characteristics function (c.f) is given as if $\psi_x^{(t)} = \left(\frac{\lambda}{\lambda - it} \right)^r$
(ii) Calculate the mean and variance if r is the cube of 12 and λ is the cube root of the sum of odd numbers between 18 and 24.
- b. Define the following terms;
(i) Level of significance (ii) Critical region
(iii) Point estimation (iv) One tail test

Question Four:

- 4a. An experiment was conducted on computer overheating problem, a random sample of 60 computers were tested for overheating issues and a mean reading of heating was recorded to be 76 and variance 21. An expert engineer have claim that the computers in the faculty Lab overheats an average of 90 when used. Test the hypothesis to show if there is a significant difference between the engineer's claim and the experimental result *at 5% Level of Significance.*
- b. State three properties of t-distribution function

Question Five:

5a. Given that x and y are two random variables, define covariance (σ_{xy}).

b. From your knowledge of covariance theorems, complete the following:

- (i) If x and y are independent random variables, $\text{cov}(x, y) = \text{-----}$
- (ii) If a and b are constants and x, y are two r.v's then $\text{cov}(ax, by) = \text{-----}$
- (iii) If c is a constant and x is any r.v, then $\text{cov}(c, x) = \text{-----}$

c. A C.P.U marketer submit several micro chips to a Laptop producer. The producer wants to know if the several micro-chips average-breaking strength exceeds 400 psi. From experience, it is known that the variance of breaking strength is 200 (psi)². In testing the hypothesis, a sample size of four taking from the several micro-chips yielded 200, 212, 215, 218 psi. Test the hypothesis at $\alpha = 0.05$.

Question Six:

6a. Differentiate between type I error and type II error

b. The joint density function of two random variable x and y is given by

$$f(x, y) = \begin{cases} \frac{xy}{100}, & 0 < x < 3, 1 < y < 4 \\ 0, & \text{Otherwise} \end{cases}$$

Find $E(3x + 3y)$

Question Seven:

7a. Suppose that a game played with a simple die (assume the die is fair). In this game a player wins N60.00 if a 2 turns up, N80.00 if a 4 turns up; and loses N50.00 if a 6 turns up; while he neither wins nor loses if any other face turns up. Find the expected sum of money won.

b. Use the method of maximum likelihood estimation to show that in a Poisson distribution, $\hat{\theta} = \bar{X}_n$ is the maximum likelihood.

Paper 2 of 2 — STATISTICAL THEORY IV STA 321 HND1 (20172018) SECOND SEMESTER

Level HND1 • Second Semester • Handout

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**DELTA STATE POLYTECHNIC,
OTEFE-OGHARA**
DEPARTMENT OF STATISTICS
SCHOOL OF APPLIED SCIENCE AND TECHNOLOGY
2ND SEMESTER EXAMINATION, 2017/2018 SESSION

Course: STA 321 – Statistical Theory IV

Instruction: Attempt any five questions

Time: 3hrs

- 1a. When do we say that two variables are stochastically independent? (4 mks)
- b. Given the joint probability distribution of x and y :

$x \backslash y$	0	1	2	3
0	0	$3/84$	$6/84$	$1/84$
1	$4/84$	$24/84$	$12/84$	0
2	$12/84$	$18/84$	$0/84$	0
3	$4/84$	$0/84$	$0/84$	0

- (i) Find the marginal distribution of x and y (3 mks)
- (ii) Are x and y independent? (5 mks)

- 2a. Let $f(a, b) = \begin{cases} 6ab^2, & 0 \leq a \leq 1, 0 \leq b \leq 1, \\ 0, & \text{otherwise,} \end{cases}$
Verify the independency of a and b . (6 mks)

- b. Let $f(x, y) = \begin{cases} \frac{2}{3}(x+2y), & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$

State whether x and y are independent. -- (6 mks)

- 3a. What is covariance? (2 mks)
- b. State four properties of covariance (1 mk each)
- c. Given two discrete random variance x, y with the joint pmf as in the table below. find the cov(x, y). (6 mks)

$x \backslash y$	-1	0	1	
-1	$1/8$	$1/8$	$1/8$	$3/8$
0	$1/8$	0	$1/8$	$2/8$
1	$1/8$	$1/8$	$1/8$	$3/8$
	$3/8$	$2/8$	$3/8$	1

- 4a(i) What is characteristic function? (3 MKS)
 (ii) State two properties of characteristic function. (3 MKS)
 b. Prove the assertion that the characteristic function of random variable x always exists. (4 MKS)
 c. (i) State the characteristic function of a Gamma distribution. (2 MKS)
 5a (i) State Gamma function (2 MKS)
 (ii) If x has a Gamma distribution with parameter r and λ . Given that $r = 12$ and λ is the square of r , find the (a) mean and (b) variance (5 MKS)
 b(i) State the Chi-square distribution (2 MKS)
 (ii) Let x be χ^2_{15} , obtain the prob $(3.25 \leq x \leq 40.5)$ (3 MKS)
 6a (i) What is F - distribution (1 MKS)
 (ii) State four uses of F - distribution (2 MKS)
 (iii) Using the formula of the mean and variance of F distribution, where $n = 3$, $m = 6$, compute the mean and the variance? (3 MKS)
 b(i) Explain critical or Rejection region with the aid of a diagram (2 MKS)
 (ii) What is a type I error and type 2 error? (4 MKS)
 7a. Define the following
 (i) Hypothesis (2 MKS)
 (ii) Null and Alternative hypothesis (4 MKS)
 b. It is known from the past experiment that the mean breaking strength of a specie of rock is 1900N and variance is m^3 given that $m = 10$. In a geological survey a random sample of 60 pieces of rock was tasted and a mean strength of 1950N was obtain. Test the Hypothesis that the population mean strength of the rock is 1900N at 5% level of significance. (6 MKS)

$$\mu = 1900 \quad 1000 \quad n = 60$$

$$\bar{x} = 1950$$

$$\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim z_{\alpha/2}$$

$$\frac{1}{2^{n/2} \Gamma(n/2)} \cdot \frac{2n^2(n+m-2)}{m(m-2)^2(n-4)}$$

$$= \frac{2(3^2)(3+6-2)}{6(6-2)^2(3-4)}$$

$$= \frac{3(7)(7-2)}{6(4)^2(-1)^2} = \frac{21(7)}{6(16)(-1)} = \frac{18.7}{-96} = -1.97$$